Indian Institute of Technology Dharwad
Department of EECE

EE 601L: Neural Networks and Deep Learning Lab

LAB-1 **January 13, 2026**

1. Implement, train, and evaluate a multi-class Perceptron classifier for a simplified version of the Iris dataset. The goal is to understand the functionality of a Perceptron, explore its training dynamics, and visualize decision boundaries for a multi-class classification problem. You will analyze the effect of the number of training iterations on the model's accuracy for both training and testing datasets.
    a. Load the Iris dataset. Extract only the first two features for 2D visualization. Split the data into training and testing sets using an appropriate train-test split ratio. Standardize the features for better convergence of the Perceptron model. Implement Multi-Class Perceptron Classifier
    b. Develop a Perceptron-based model capable of handling multi-class classification tasks. Train the model using a defined learning rate and a configurable number of iterations. Update weights based on Perceptron training rules to handle multi-class predictions. Training and Evaluation:
    c. Train the Perceptron model with varying numbers of iterations (e.g., 10, 50, 100, and 200). Evaluate the accuracy of the model on both training and testing datasets for each iteration setting. Compare the accuracy trends for different numbers of iterations. Visualization:
    d. Plot accuracy trends for training and testing datasets as a function of the number of iterations. Visualize decision boundaries for the trained Perceptron model across different iteration values to understand how decision regions evolve with training. Analyze and Conclude:
    e. Interpret the results by comparing training and testing accuracies for different iteration settings. Discuss the trade-off between iteration count and model performance. Comment on the limitations and potential improvements of the Perceptron algorithm for multi-class classification.

Deliverables:

Python code for all the tasks mentioned above. Plots showing accuracy trends and decision boundaries. A report summarizing your analysis, observations, and conclusions.